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FIRST SEMESTER B. TECH EXAMINATION, JANUARY - 2024
ENVIRONMENTAL ENGINEERING & SUSTAINABILITY

Time: 3 Hours **Maximum Marks: 100**

Instructions:

i. Missing data may be suitably assumed.

ii. Answer all the questions from Part – A, any 6 questions from Part – B & C

iii. Draw neat sketches wherever necessary

		PART-A	**05 X 02=10**		
1.	**a**	Define the following terms: **i.** Osmotic pressure **ii.** Reverse Osmosis	4	03	**02**
	b.	What are Green Electronics? Give some examples of favorable materials used in the manufacture of green electronics.	3	02	**02**
	c.	Name any four best electronic water purifiers available in the market.	4	03	**02**
	d.	Define e-waste? what are the major components of e-waste management system.	5	04	**02**
	e.	Differentiate between Distillation & Demineralization.	4	01	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Write a short note on renewable energy sources.	2	01	**05**
		OR			
	b.	Write a note on Hydrogen Fuel cell. What are the technical challenges associated with the storage of Hydrogen.	2	01	**05**
3.	**a.**	List the properties & benefits of Aluminium.	3	02	**05**
		OR			
	b.	Write a short note on Eco-friendly laptops.	3	02	**05**
4.	**a.**	Explain the steps involved in the working of Solar powered water filtration system.	4	03	**05**
		OR			
	b.	Explain the working principle involved in the water purification by Activated Carbon.	4	03	**05**
5.	**a.**	Write a short note on the steps involved in the e-waste recycling process.	5	04	**05**
		OR			
	b.	Explain CRT (cathode ray tube) Recycling process.	5	04	**05**
6.	**a.**	What are the responsibilities of SPCB's in pollution control.	5	05	**05**
		OR			
	b.	Write a short note on AIR (prevention & control of pollution) act	5	05	**05**

		1981.			
7.	**a.**	Explain the working principle of Reverse Osmosis & also list its advantages.	4	03	**05**
		OR			
	b.	Explain water purification by ION Exchange process	4	03	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	What are the advantages & disadvantages of Renewable energy.	2	01	**10**
		OR			
	b.	Explain wind conversion technologies.	2	01	**10**
9.	**a.**	Explain the applications of the following **i.** Borosilicate glass **ii.** Graphene	3	02	**10**
		OR			
	b.	Discuss about Iron alloys & their importance	3	02	**10**
10.	**a.**	Explain Water Purification by UV radiations. Describe the points to be considered for the maintenance of UV water treatment system	4	03	**10**
		OR			
	b.	Write a short note of Ultra filtration water purifier & also differentiate between ultraviolet & ultra filtration purifiers.	4	03	**10**
11.	**a.**	Explain the classification & Composition of e-waste in Indian Scenario.	5	04	**10**
		OR			
	b.	Explain the challenges of E-Waste management in India, also explain the Practices for efficient E-Waste management.	5	04	**10**
12.	**a.**	Write a short note on Water (Prevention & control of pollution) act 1981 & Plastic waste management rules 2016	5	05	**10**
		OR			
	b.	Explain the salient features of the following **i.** Wild life protection policy **ii.** Environment protection act 1986	5	05	**10**
13.	**a.**	Explain about sediment filters & list the benefits of portable solar power water filtration systems	2	01	**10**
		OR			
	b.	What is Solar water disinfection? How does it work & also explain the advantages of solar water disinfection.	2	01	**10**